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Translating CBCT to CT

Using 3d CycleGAN

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Abstract

This report documents our process of developing a model capable of translating between CBCT and CT scans. After a review of the available literature on the subject, the model of choice is the 3d CycleGAN. The individual parts of the CycleGAN, i.e. the discriminator and generator, are based on already existing code that is adjusted and sometimes simplified for our purposes. While our original goal of developing a model to be used by Rigshospitalet in their Adaptive Radiation Therapy program was not met, the developed CycleGAN is demonstrated to be able to learn. Furthermore, two CycleGANs are trained on downsampled volumes and patches, respectively, and are demonstrated to translate in very different manners with different strengths and weaknesses. Finally, it is discussed in which ways the model and the training of it could be improved in order to be able to reach the original goal.

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1 Introduction

In the course of radiotherapy, the target area and the surrounding anatomy will most likely undergo changes. The tumour might change size, the patient may have lost or gained weight, the bladder may be more or less full, etc. [Dong et al., 2021]. The planning computed tomography (pCT) scan taken at the outset of the treatment and used to calculate the radiation dosage thus may differ significantly from reality on the day of radiotherapy, resulting in inaccurate dosage calculations. Adaptive radiation therapy (ART) seeks to solve this problem by e.g. making new scans in real-time with up-to-date anatomical information [Yan et al., 1997]. To avoid too much radiation dosage to the patient, CBCT rather than CT scans are being used. While they require lower radiation dosage and therefore are a lower risk for the patient, CBCT scans are of much lower quality with their imaging artifacts and increased noise, making it not very suitable for dosage calculation [Dong et al., 2021]. At present, one way to resolve this is by registering the planning CT scan with the CBCT scan to create the deformed planning CT (dpCT) scan, which attempts to preserve the real Hounsfield unit values of the CT while adjusting to the anatomical changes provided by the CBCT. However, as this system is not perfect, there is interest in developing a method independent of registration against which to check the system (Petersen, J. (personal communication, June 11,2021)). Of particular interest is a machine learning model that could translate the real-time CBCT scan into a synthetic CT (sCT) scan for radiotherapists to compare with their registered dpCT scan.

This report concerns our progress towards developing such a model. Although our time frame did not allow us to reach this goal, in the course of the project we develop a working 3d CycleGAN - chosen on the background of a literature review - deal with the different challenges of the data preprocessing steps, and demonstrate some of its capabilities as well as discuss ways in which to improve the model.

2 Literature review

Deep-learning based medical image translation can be challenging due to the volumetric nature of medical images, with GPU memory consumption being a large roadblock. Many well-functioning image translation methods exist for 2d images, and can also be applied to 2d slices of a 3d image, but this can be cause for issues such as discontinuities between the slices and degradation of image quality. The challenge is thus to find a method that processes the training data in a way such that learned feature/activation maps can fit into GPU memory and at the same time produce images of good quality / preserve the quality of the 3d image being translated. Nowadays, generative adversarial networks (GANs) are among the most popular models used for many medical applications and other computer vision tasks, since they can learn to synthesise practically any type of image without the need for

paired/labeled data. In this review, we dive deeper into the recent literature on GANs and other architectures used for medical applications, aiming to settle upon a suitable model and architecture for our particular problem of translating between the image modalities CBCT and CT.

2.1 GANs

Medical image synthesis with GANs can be broadly categorized into unconditional and conditional image synthesis. The GAN is originally an unsupervised, unconditional generative model, which generates new data given random noise, leaving the user with no control on the actual data generation. [Kazeminia et al., 2020].

2.1.1 The GAN framework

The GAN framework was originally designed by [Goodfellow et al., 2014]. A GAN consists of two competing deep CNNs, a generator and a discriminator, that are trained simultaneously. A sample, e.g. an image, is input to the discriminator, which outputs the probability of the sample being real. The discriminator is essentially a binary classifier, which outputs 1 if the sample is real, and 0 if it is fake. The generator is input samples from a known simple distribution e.g. uniform and transforms this random input/noise vector z into a fake sample/image y. The generator aims to find a mapping $y = G(z)$ from z to the real data. The goal is then to optimize this mapping such that the distribution of the generated data Y resembles the distribution of the training data X. During training the generator is penalized (generator loss) for generating a sample that the discriminator classifies as fake. It thus gets better at synthesizing more realistic images, while the discriminator gets better at discriminating the real from the synthesized images. This is called adversarial training. In mathematical terms, the generator tries to minimize the following cross-entropy loss function while the discriminator tries to maximize it [Kazeminia et al., 2020], [Wei et al., 2018]:

$$\min_G \max_D V(D,G) = E_x[logD(x)] + E_z[log(1 - D(G(z)))] \tag{1}$$

where $D(x)$ is the discriminator's output probability of a real sample x being real, E_x is the expected value over all real samples, $G(z)$ is the fake sample output by the generator, $D(G(z))$ is the discriminators output probability of the fake sample being real (hence $1 - D(G(z))$ is the probability of a fake sample being fake), and E_z is the expected value over all fake samples. The generator cannot directly affect the $logD(x)$ term of the function, so instead it minimizes the $log(1 - D(G(z)))$ term i.e. the log probability of fake sample being fake (thus tries to fool the discriminator).

2.1.2 Unconditional GANs

An example of an unconditional GAN is the DCGAN, introduced in [Radford et al., 2015]. The model has e.g. been used to synthesize high quality focal liver lesion patches from CT images for the specific task of liver lesion classification [Frid-Adar et al., 2018]. Here, input to the generator was a vector of 100 random numbers drawn from a uniform distribution, and output was a liver lesion image of size $64 \times 64 \times 1$, which was then fed into the discriminator to be classified as real or fake. The authors trained a separate DCGAN for each class of liver lesion (cyst, metastasis, hemangioma).

The DCGAN was also used to synthesize CT images from CBCT images of head-and-neck cancer patients for use in adaptive radiation therapy [Liang et al., 2019].

We will not discuss unconditionals GANs further, since most studies have employed conditional GANs for image-to-image translation tasks.

2.1.3 Conditional GANs

The conditional GAN (cGAN), originally proposed in [Mirza and Osindero, 2014], incorporates additional prior information like class labels in the image generation process. It was shown that conditioning the GAN on prior knowledge both stabilizes the training and improves the generation of detailed features in the target image.

The authors of [Isola et al., 2016] proposed a variant of the cGAN model called Pix2Pix, for high resolution image-to-image translation. In contrast to the original GAN framework, the Pix2Pix model requires paired data and is thus a supervised framework.

CycleGAN introduced by Zhu et al. (2017) [Zhu et al., 2017] can be used to transform images between two domains. This framework combines two GANs to find a mapping from domain X to domain Y and vice versa. The GANs are chained together, and a cyclic loss function tries to minimize the difference between the original and reconstructed image.

2.1.4 Challenges in training GANs

Training a GAN can pose several problems. In the adversarial training, the parameters of the generator and the discriminator are updated alternately using gradient-based optimization methods. The generator is updated only through gradients back-propagated from the discriminator. Convergence is achieved when the discriminator cannot distinguish between real and fake samples, i.e. when Nash equilibrium (i.e. saddle point) is found. The generator and discriminator may evolve together, but there is no guarantee of balance between the training of the generator and discriminator with the Jensen-Shannon (JS) divergence, which is minimized by optimizing for the adversarial loss function (see detailed description of the GAN

framework). Consequently, one network may be more powerful than the other, which in most cases is the discriminator. When the discriminator becomes too strong as opposed to the generator, it gets to easy for the discriminator to discriminate between the real and fake samples, which results in *vanishing gradients*, and there is nothing to guide the training of the generator. Another common problem is *mode collapse*, where the GAN map all input samples to very similar synthetic samples covering only a single or few limited modes of the data distribution, and the optimization thus stops to make progress [Yi et al., 2019], [Zhu et al., 2017]. Several variants of GANs has been proposed to address these problems, including some of the models reviewed here, but we will not get into the exact details of such mechanisms here as it is not our focus.

2.1.5 Cross-modality synthesis using cGANs

2d models Many different methods have been proposed for cross-modality image synthesis. E.g. MedGAN [Armanious et al., 2018] is a framework for medical image translation. It extends the Pix2Pix framework (a variant of the conditional GAN) [Isola et al., 2016] with a cascaded U-net generator architecture and additional non-adversarial losses, such as perceptual loss and style transfer loss (calculated as the differences between feature maps of the translated images and the ground truth images). The generator architecture is called CasNet, which concatenates several fully convolutional encoder-decoder networks with skip connections (U-blocks) into a single generator network in an end-to-end manner. The cascaded architecture progressively refines the translated images. Besides the CasNet, the framework comprises a discriminator, whose architecture is a modified PatchGAN, and a pre-trained feature extractor. In contrast to a classic GAN discriminator, which classifies the synthetic and real images as being real or not, the PatchGAN is designed to have a smaller receptive field such that it classifies smaller image patches and then averages out the result. Since the attention of the discriminator is limited to smaller image patches, it motivates the generator to produce more detailed translated outputs [Armanious et al., 2018]. The pre-trained feature extractor is used to extract deep rich features from a larger receptive field to calculate style transfer losses in order for the translated image to match style, textures and content of the target image.

However, the MedGAN framework is a supervised method for 2d images and was not yet appropriately adapted to operate on 3d medical volumes at the time of release (2018). In 2019, Armanious et al. [Armanious et al., 2019] proposed the Cycle-MedGAN framework, which expands the CycleGAN by adding two non-adversarial losses analogous to the perceptual and style transfer losses of MedGAN. Unlike MedGAN though, the calculation of such losses does not require paired data, but instead the penalized deviation is between the input images and the cycle-reconstructed images. The authors thus introduce the cycle-perceptual loss, calculated for both generators as the MAE between intermediate feature maps extracted

by the pre-trained feature extractor network (BiGAN discriminator [Donahue et al., 2016]). Additionally, they introduced the cycle-style loss, which aims to match the style (texture) of the input image onto the cycle-reconstructed images. They compared their framework to Cycle-GAN and UNIT (described later) on MR motion artifact correction and PET to CT translation. Qualitatively, the proposed Cycle-MedGAN out-performed the other two frameworks on all chosen metrics. E.g. the translated images produced by UNIT were blurry and distorted, whereas the images produced by CycleGAN matched the global structure of the ground-truth images but did not translate finer details such as bone structures precisely. In comparison, the Cycle-MedGAN produced sharper and more agreeing bone structures.

CycleGANs have shown to successfully transform 2d MR image slices into 2d CT image slices, or vice versa, without the need for co-registered pairs, e.g. by [Wolterink et al., 2017] and [Chartsias et al., 2017], respectively. The authors of the former used MR and CT images of the brain, and images of the same patient were aligned using rigid registration to allow for voxel-wise comparison of synthesized and reference CT images. They found that training using unpaired images (MR/CT training images were of the same patient set, but given randomly in an unpaired manner) was even better than using aligned images as the mapping is not affected by co-registration artifacts. Chartsias et al. used a cardiac dataset of MR and CT images coming from different patients, and since they could not evaluate the synthetic images directly (no ground truth images existed), they utilized them to achieve improved results in segmentation. They included both a CT image and its segmentation mask as two channel inputs to the CycleGAN, thus training the model to be able to transform CT images and their ground truth masks into synthetic MR images with corresponding synthetic masks.

A 2d CycleGAN was also used to synthesize CT images from CBCT images of head-and-neck cancer patients [Liang et al., 2019]. The goal was that synthesized CT images kept the CBCT anatomy while having the HU accuracy of the deformed planning CT (dpCT) (which keeps the same HU accuracy as CT). The CT images in the training dataset were re-sampled to the same resolution as CBCT and aligned with the CBCT images, and the training images were given in a randomized unpaired way. Synthetic CT images (sCT) were benckmarked against dpCT (obtained using the DIR method) with regard to the HU value accuracy. A U-Net architecture was used for the generators and PatchGAN for the discriminators. The authors quantitatively compared CBCT to dpCT, and sCT to dpCT using MAE, RMSE, SSIM and PNSR to evaluate the HU accuracy in all test images. They also compared the performance of CycleGAN against the DIR method in terms of anatomical accuracy in a phantom study, where it was possible to acquire CBCT and CT images with exactly same anatomy. Results showed improved HU accuracy of sCT images (generated by CycleGAN from CBCT) compared with CBCT images, while anatomy of CBCT is preserved in sCT images and more accurate than dpCT. The also compared the CycleGAN model to the DCGAN ([Radford et al., 2015]) and progressive growing of GANs ([Karras et al., 2017]) with the same training

and testing dataset, and CycleGAN outperformed these two models with respect to best image quality both visually and quantitatively.

Another GAN framework for unpaired cross modality synthesis is the UNIT model, which is an unsupervised translation framework that maps the corresponding images in two different domains into a common latent space using a pair of Variational Auto Encoder-GANs (VAE-GANs) [Liu et al., 2017]. Each VAE is composed of an encoder-generator pair. One VAE, say VAE_1 maps an image of domain A to a code in a shared latent space via its encoder E_1 and then decodes a random-discomposed version of the code to reconstruct the image in domain B via its generator G_1. The other VAE, VAE_2, then does the same thing on an image of domain B, which is then translated to an image from domain A. In a study by [Welander et al., 2018], it was found that UNIT and CycleGAN (2d) performed almost equally well for the transformation between T1 and T2-weighted MR images of the brain. The authors suggested that 3D GANs can potentially yield even better results (see the '3d models' section).

A common problem with CycleGAN is that it does not guarantee consistent anatomical structures in the input and synthetic images [Yang et al., 2020], [Zhang et al., 2018]. One could imagine a scenario in which one generator expands and distorts the anatomical geometry a bit, while the other generator shrinks and de-distorts the geometry. This is allowed, since cycle-consistency is maintained and the images appear valid, but obviously very undesirable because the anatomy is not preserved in the synthetic images. The authors of [Yang et al., 2020] proposed a Cycle-GAN model for unsupervised brain and abdomen MR-to-CT synthesis method that constrains structural consistency between synthetic and input images without additional requirements on the training dataset such as ground-truth segmentations (see description of study by [Zhang et al., 2018] below). They define an extra structure-consistency (sc) loss as the voxel-wise difference between modality independent structural features (MIND features) in the input and synthetic images. The structural-consistency loss between input and synthetic images is supposed to guide the training process of generators. Their networks were 2D and performed on sagittal slices. They compared different unsupervised methods for the brain and abdomen MR-to-CT synthesis on a brain dataset and the CHAOS dataset [Emre Kavur et al., 2020], respectively, and the results showed that their sc-CycleGAN performed better than both the CycleGAN and gc-CycleGAN (gradient-consistency loss proposed in [Hiasa et al., 2018], regularizing the loss function in order to improve accuracy at the edges/boundaries of the synthesized images). The authors also investigated the impact of training dataset setting on the performance of unsupervised synthesis methods. They compared the results obtained when using a paired training dataset and from unpaired setups. In the paired setting, pre-acquired non-rigidly aligned MR-CT pairs (from same patients) were used for training. In the unpaired setups, the authors used MR and CT volumes that respectively had and had not been rigidly aligned as pairs before the randomized unpaired selection of MR and CT samples during training. Paired training data produced the

best results, but affine pre-registration of the unpaired data improved the synthesis performance compared to unpaired data without registration.

3d models The authors of [Zhang et al., 2018] also tried to overcome the problem of shifted anatomy in the synthetic images by proposing a method to realize both medical volume translation as well as segmentation with unpaired 3D heart/-cardiovascular data. They used a 3d CycleGAN to illustrate the key idea and employ an additional shape consistency loss that is obtained from/supervised by a segmentation network, which is supposed to mitigate the geometric distortions in the transformation that may destroy anatomical structures. Their model is composed of several 3D CNNs. However, their method requires a training dataset with ground-truth segmentations, which may be difficult to obtain.

In a supervised setting, a method to translate 3d MR to CT images was proposed by [Nie et al., 2017]. The authors used two datasets to test their method, a brain dataset and a pelvic dataset. Their proposed GAN model is composed of a cascade of Generator networks (here as 3D Fully Convolutional Networks). The network was trained on patches that contain only local information, which means that the network learns less of the global context. To compensate for this, they used the Auto-Context Model (ACM), which involves iterative training of several GANs for making the model context-aware. Each GAN took as input MRI patches and predicted corresponding CT patches, which were then concatenated with the MRI patches as two channel inputs to the next GAN. At testing, an MR image was first partitioned into overlapping patches and input into the generator, which generated the corresponding CT patches that were iteratively refined through the cascade of 3D FCNs and finally merged into a single CT image. The results showed improved performance by the use of ACM, and the proposed GAN approach out-performs a traditional FCN (i.e. without adversarial training) with less artifacts in the generated data etc. However, the translation method by Nie et al. requires corresponding pairs of CT and MR images of the same patient, which are difficult to obtain.

In contrast, a CycleGAN can be trained using unpaired data, as mentioned previously. In a study by [Pan et al., 2018], a 3D CycleGAN was proposed for synthesizing PET from MR images for Alzheimer's Disease Diagnosis. The authors evaluated the proposed method on two subsets of the ADNI database [Jack Jr et al., 2008], including ADNI-1 and ADNI-2. The architecture of each Generator composes an encoding part, a transferring part composed of residual network (ResNet) blocks, and a decoding part. Each discriminator contains five convolutional layers. The averaged peak signal-to-noise ratio (PSNR) was used to measure the image quality of the synthetic PET images generated by their proposed method. The trained model showed reasonable results, and the synthetic PET images had acceptable image quality in terms of PSNR.

Abramian and Eklund (2019) also used a 3D CycleGAN, but instead for synthesizing fMRI volumes from T1-weighted volumes and vice versa. Their CycleGAN

model is based on the 2d CycleGAN implementation by [Welander et al., 2018]. They replaced all 2d convolutions by 3d convolutions and made other changes as well. The model was trained on fMRI and T1 volumes of the brain from the 1000 functional connectomes project [Biswal et al., 2010] with focus on the Beijing and Cambridge datasets. Due to varying image sizes and voxel sizes, they initially registered the fMRI and T1 data to the MNI152 brain template to facilitate training and evaluation. To reduce training time and memory consumption, the MNI152 1 mm brain template was cropped from 182 x 218 x 182 to 152 x 180 x 120 voxels. The authors focused on transforming a full volume, rather than working on subvolumes/patches, which can reduce the performance due to the lack of context information for each training sample. They showed that the fMRI-T1 transformation was rather uncomplicated for slices in the upper part of the brain, but was more challenging for slices in the lower part of the brain due to severe distortions.

The authors mentioned above (D. Abramian and A. Eklund) also participated in a study [Cirillo et al., 2020] to demonstrate the use of a 3d GAN for brain tumor segmentation. Their model is called Vox2Vox and is based on the Pix2Pix GAN ([Isola et al., 2016]) modified to 3d, and showed promising results.

In a more recent study, Sun et al. translated from CBCT to pseudo-CT (synthetic CT) images using a 3d CycleGAN with residual layers, attention gates and addition of a gradient loss function in relation to the generator [Sun et al., 2021]. Attention gates have been used to highlight important image features and suppress irrelevant regions such as artifacts in the CBCT images. In the generator (U-Net network), a gating signal was added before each skip connection to the output of the encoder and corresponding decoder under each resolution operation. Furthermore, in the study, a gradient loss function was added to obtain as accurate texture information in the synthesized images as possible. For training and prediction, they used 120 sets of CT-CBCT image pairs obtained from 120 different abdominal cervical cancer patients. The authors performed registration based on bone landmarks on the CT-CBCT image pairs to facilitate the training, even though this was not necessary for the CycleGAN model. They trained the model on overlapping 32 x 32 x 32 patch blocks for the convenience of GPU memory.

The synthesized CT images were compared to synthetic CT images generated with a 3D GAN with U-net generator (unet-GAN) and FCN generator (FCN-GAN). The accuracy difference between synthetic CT and real CT images was evaluated using voxel-wise mean absolute error (MAE) and structural similarity coefficient (SSIM) during a fivefold cross-validation technique. Results showed that the synthetic CT images synthesized by the proposed CycleGAN model were closer to the real CT images (in HU values for MAE, and anatomical structure) with higher quality, compared to a those obtained with 3D unet-GAN and 3D FCN-GAN. To prove the effect of the gradient loss function on CycleGAN, the used both a CycleGAN model with and without gradient loss in the CT synthesis. They found that the quality of the synthetic CT images obtained with a CycleGAN with gradient loss

was significantly better, both in terms of details and anatomical structure.

In the study, the synthetic CT images obtained by 3d CycleGAN and 2d CycleGAN were also compared. The 3d model generated images that were more continuous in depth, and the 3d method was in general more accurate than the 2d method.

Other models Zeng et al. (2019) [Zeng and Zheng, 2019] introduced a hybrid GAN (hGAN) consisting of a 3D generator network (3D FCNs) and a 2D discriminator network for brain MR to CT synthesis using unpaired data. They chose a 3D generator in attempting to solve the spatial inconsistency problem that arises when constructing a 3d image from slices generated with a 2d model. The architecture of the 3d generator composes an encoding and decoding part with residual blocks in-between. The authors chose to use a 2D discriminator to address the issue of lacking context information for each training sample when training a 3d discriminator on smaller image patches that contains only local information. The discriminator is composed of convolutional layers. During training, the 3d generator was weakly supervised by comparing its output to the corresponding weak label volume (using L1 loss) that was constructed from a sCT image generated from a trained 2D CycleGAN. The generated 3d synthetic CT patch was then converted to a batch of 2d slices, which was then used together with 2d slices from an unpaired real CT image as input to the 2d discriminator to be classified as real or fake. The authors compared their method to 2D CycleGAN [Zhu et al., 2017] and 3D CycleGAN [Zhang et al., 2018] with a standard 2-fold cross-validation technique. Their results demonstrated that their proposed hGAN achieved better results than both the 2D CycleGAN and 3D CycleGAN when only limited number of unpaired data were available. The hGAN outperformed the other models in all metrics (MAE, relative MAE, and PSNR), and visually, the proposed hGAN showed better results in terms of spatial consistency and smoothness. However, their method could not guarantee structural consistency between the input and synthetic images, because CycleGAN has no direct constraints between these.

2.2 Non-GAN based methods

The authors of [Han, 2017] were among the first to develop a method for synthetic CT generation from MR based on a deep CNN (convolutional neural network). Their 2D CNN was trained by minimizing the voxel-wise differences between the synthesized CT volumes (generated from MR volumes that were processed slice-by-slice) and reference CT volumes. It was a requirement that the reference CT images were rigidly aligned with the input MR images, otherwise slight voxel-wise misalignment of MR and CT images could result in blurry synthetic images. As mentioned earlier, Nie et al. [Nie et al., 2017] addressed this image misalignment issue by combining the voxel-wise loss with the image-wise adversarial loss in a GAN, but their approach still has a voxel-wise loss component that requires paired MR and CT volumes for training. However, since then, GAN-based methods have

been the superior choice for medical image translation tasks, which is why GANs are in focus for our literature review.

2.3 Discussion of 2D vs. 3D models/CNNs

Below is an overview of the methods from our literature review.

Publication	Modalities	Method	2D/3D	Supervised/ unsupervised	Body part
Frid-Adar et al. (2018)	CT	DCGAN	2D	unsupervised	liver
Armanious et al. (2018)	PET ↔ CT	MedGAN (Pix2Pix)	2D	supervised	brain
Armanious et al. (2019)	PET ↔ CT	Cycle-MedGAN	2D	unsupervised	brain
Wolterink et al. (2017)	MR ↔ CT	CycleGAN	2D	unsupervised	brain
Chartsias et al. (2017)	CT ↔ MR	CycleGAN	2D	unsupervised	heart
Liang et al. (2019)	CBCT ↔ CT	CycleGAN	2D	unsupervised	head and neck
Welander et al. (2018)	T1 ↔ T2 MR	CycleGAN, UNIT	2D	unsupervised	brain
Yang et al. (2020)	MR ↔ CT	sc-CycleGAN	2D	unsupervised	brain, abdomen
Hiasa et al. (2018)	MR ↔ CT	CycleGAN	2D	unsupervised	musculoskeletal
Zhang et al. (2018)	CT ↔ MR	CycleGAN	3D	unsupervised	heart
Nie et al. (2017)	MR → CT	Cascade GAN	3D	supervised	brain, pelvis
Abramian and Eklund (2019)	T1 MR ↔ fMRI	CycleGAN	3D	unsupervised	brain
Sun et al. (2021)	CBCT ↔ CT	CycleGAN	3D	unsupervised	abdomen
Pan et al. (2018)	MR ↔ PET	CycleGAN	3D	unsupervised	brain
Zeng et al. (2019)	MR ↔ CT	hybrid GAN	3D/2D	unsupervised	brain

We have seen that for 3d image translation there are several different options. Frameworks such as CycleGAN, which was initially proposed for 2d images, can be trained and then applied to every slice of a volume as done by [Wolterink et al., 2017] and many others. However, this approach does not take advantage of the 3d spatial information and can lead to discontinuity across slices [Zeng and Zheng, 2019], [Abramian and Eklund, 2019], [Nie et al., 2017]. A 3d CycleGAN on the other hand can better model the spatial information and thus solve the discontinuity problem. However, training a 3d CycleGAN has higher computational cost since 3d convolutions involves more calculations than 2d convolutions, and high GPU memory consumption because of the activations for each layer in each CNN (in CycleGAN there are four CNNs trained simultaneously) that has to be stored in order to calculate gradients using backpropagation [Abramian and Eklund, 2019], [Gomez et al., 2017]. Training a 3d network on subvolumes/patches instead of full size volumes substantially reduces memory consumption, but may reduce performance as well, since the patches only contain local information and thus lack global context. [Nie et al., 2017] used a 3d cascade GAN to iteratively refine the output of the generator, thus making it context-aware, while [Sun et al., 2021] tried other approaches such as the addition of a gradient loss and attention gates in order to improve their 3d CycleGAN. [Zeng and Zheng, 2019] proposed a 3D/2D hybrid GAN with a 3D generator to better model the spatial information and a 2D discriminator to solve the lack of global context problem, because a 2d discriminator can take complete slices as the input. E.g. a comparison of synthetic CT images generated from MR with different methods can be seen in the study by [Zeng and Zheng, 2019] as an example 1.

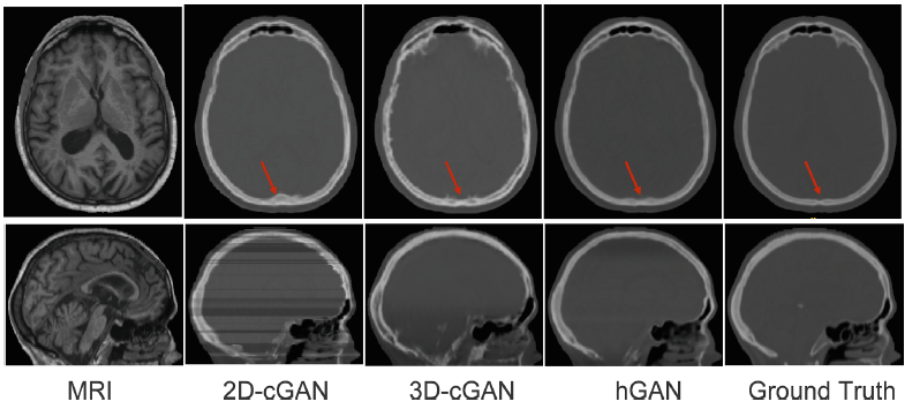


Fig. 3. Visual comparison of sCTs generated with different methods.

Figure 1: from [Zeng and Zheng, 2019], p. 764

In the study, the spatial inconsistency is pronounced in the image generated using a 2d CycleGAN (2d-cGAN in the figure) with black horizontal lines across the image. Their proposed hGAN produces more smooth images compared to the 3d CycleGAN (implemented based on the paper by [Zhang et al., 2018]), but the authors remark that the performance of the 3d-cGAN could probably be improved with more data.

To deal with the issues of memory usage caused by moving to 3D, Uzunova et al. [Uzunova et al., 2020] proposed a multi-scale patch-based GAN approach for memory-efficient image generation in unpaired domain translation. The idea is to first generate a low-resolution version of the image using an LR (low resolution) GAN, then afterwards increase the resolution by using multiple HR (high resolution) GANs to generate constant-size patches which are conditioned on the previous resolution scales. In this way, patch artifacts are avoided, global information is included and memory requirements remain constant independently from the image size. The authors of the study had focus on unpaired topology preserving domain translation and used edge information as a topology constrain. The proposed GAN proved useful for unpaired domain translation e.g. MRI T1 to MRI T2, but using edge information as topology constrain may pose a problem in translation between modalities, e.g. CT to MRI, since their extracted edges lack correspondence.

A U-Net architecture is commonly used for the generator in a GAN. U-Nets have high memory requirements, especially in 3d, as discussed previously. Several modifications to the architecture have been tried out in order to ease the pressure on GPU memory, most frequently employed in segmentation tasks ([Cheng et al., 2020], [Brügger et al., 2019], [Etmann et al., 2020]). Other 3D GAN related methods have been tried as well to alleviate the memory burden [Sun et al., 2020], but we will not discuss these further here. However, these approaches could be considered when

working with a 3D GAN.

2.4 Choice of model based on literature review

On the basis of the literature review, it seems the CycleGAN framework has been the most popular choice for cross-modality image synthesis when paired training data is unavailable, and it has achieved good results. Since we expect to work with unpaired CBCT/CT images, we choose to go with a CycleGAN model for our translation task.

Several studies employed modifications on the basic CycleGAN framework either on the network architecture or additional losses, which improved the performance of the model in relation to common problems such as discontinuity across slices or shifted anatomy in the synthetic images. Most of the implementations used in such studies are unavailable and rather complicated, so we preferred to implement the basic framework. [Sun et al., 2021] and [Zeng and Zheng, 2019] compared the performance of a 2d CycleGAN vs. a 3d CycleGAN. In both studies 3d CycleGAN out-performed 2d CycleGAN because it is better to model the spatial information in the 3d images. On this ground and out of personal interests, we choose to implement a 3d CycleGAN. In training our model, we will consider the various approaches encountered in the literature review that deal with the issues of GPU memory when moving to 3d.

2.5 CycleGAN

The CycleGAN model proposed by [Zhu et al., 2017] consists of four main components, two generators (G and F) and two discriminators (D_X and D_Y). $G : X \rightarrow Y$ generates images of domain Y based on images from domain X, and $F : Y \rightarrow X$ generates images of domain X based on images of domain Y. D_X discriminates images of domain X as real or fake, and encourages F to generate images that are indistinguishable from domain X, and similarly for D_Y and G. This happens through optimization of the adversarial loss function described previously, for both mappings. For recap, the adversarial loss for e.g. the mapping function $G : X \rightarrow Y$ and corresponding discriminator D_Y is expressed as:

$$\begin{aligned}\mathcal{L}_{GAN}(G, D_Y, X, Y) = &\ \mathbb{E}_{y \sim p_{data(y)}}[\log D_Y(y)] \\ &+ \mathbb{E}_{x \sim p_{data(x)}}[\log(1 - D_Y(G(x))]\end{aligned}$$

where G tries to minimize this function, while D_Y tries to maximize it. Similarly for the mapping function $F : Y \rightarrow X$ and discriminator D_X.

Since adversarial losses alone cannot guarantee that an individual input x_i is mapped to a desired output y_i (due to the many possible mappings that may cause the same distribution over $\hat{y}_i$), cycle-consistency is introduced to further regularize the mappings i.e. $G(F(y)) \approx y$ for each image y from domain Y and, $F(G(x)) \approx x$ for

each image x from domain X. The cycle-consistency loss is defined as:

$$\mathcal{L}_{cyc}(G,F) = \mathbb{E}_{x\sim p_{data}(x)}[\|G(F(y)) - y\|_1] \\ + \mathbb{E}_{y\sim p_{data}(y)}[\|F(G(x)) - x\|_1]$$

The total loss is then defined by the authors as:

$$\mathcal{L}(G,F,D_X,D_Y) = \mathcal{L}_{GAN}(G,D_Y,X,Y) \\ + \mathcal{L}_{GAN}(F,D_X,Y,X) \\ + \lambda\mathcal{L}_{cyc}(G,F)$$

where λ controls the relative importance of the cycle-consistency loss. The CycleGAN thus aims to solve

$$G,F = \arg\min_{G,F} \max_{D_X,D_Y} \mathcal{L}(G,F,D_X,D_Y)$$

The CycleGAN model is shown below in relation to CBCT to CT translation and vice versa. The figure is from the article by [Liang et al., 2019].

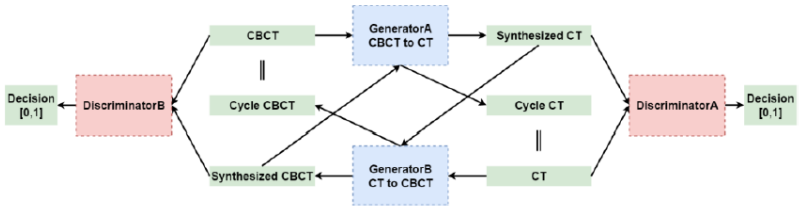


Figure 1. The CycleGAN architecture is used to convert CBCT images to sCT images.

Figure 2: by [Liang et al., 2019]

In the figure, GeneratorA is used to translate from domain B (CBCT) to A (CT), and GeneratorB is used to translate from domain A (CT) to B (CBCT). DiscriminatorA is used to classify images of domain A (i.e. real CT and synthetized CT (sCT)) as real or fake, while DiscriminatorB is used to classify images of domain B (i.e. real CBCT and synthesized CBCT (sCBCT)) as real or fake. In the first cycle, GeneratorA takes CBCT as input and generates sCT, which is then input to GeneratorB and reconstructed as cycle CBCT. At the same time, DiscriminatorA classifies real and fake CT images. In the second cycle, GeneratorB takes CT as input and generates sCBCT, which is then input to GeneratorA and reconstructed as cycle CT. At the same time, DiscriminatorB classifies real and fake CBCT images.

In the Method section we will describe the details of our 3d CycleGAN implementation, including network architecture and training details.

3 Method

3.1 Datasets

The initial plan was to start developing our model and do experiments using open data sets instead of the data sets at Rigshospitalet in order to be able to work independently at first. When we started, we were unable to find any good open CT/CBCT data sets, so we ended up using the BraTS2020 MRI data set [Menze et al., 2014]. Later, we came across an open CT-CBCT data set, namely the Pancreatic-CT-CBCT-SEG data set from The Cancer Imaging Archive [Hong et al., 2021] [Clark et al., 2013], and continued our experiments using this. There were big differences in how to handle the two data sets, which is described in further detail below and in the preprocessing section.

3.1.1 BraTS 2020 MRI

The BraTS 2020 data set consists of MRI brain scan volumes split into a training set of 367 patients and a validation set of 125 patients. For each patient, there are 4 MRI scans, namely T1, T2, Flair and T1ce scans. For our purposes, we needed to work on translating from one scan modality to another scan modality. We chose the T2 as our A domain and T1ce for our B domain for the sole reason that they were the easiest to tell apart. All scan volumes across all patients have the same format of 240 $\times$240$\times$155 (height $\times$ width $\times$ depth) and are in NIFTI format.

3.1.2 Pancreatic-CT-CBCT-SEG

The Pancreatic-CT-CBCT-SEG data set consists of 40 CT and 80 CBCT scans from 40 pancreatic cancer patients (one treatment planning CT and two CBCT scans from each patient), obtained from The Cancer Imaging Archive (TCIA). The original data is in DICOM-format, with each image composed of a varying number of DICOM files, one DICOM file corresponding to a 2d slice of the image. All 2d slices of all images have the dimensions 512 $\times$ 512 (height $\times$ width), while the number of slices (depth) in an image vary from 69 to 268 among the patients.

Because it was not clear from the organisation of the data (e.g. (sub-)directory names) which images were CT and CBCT (and it was difficult to infer this by inspection with no medical knowledge), we sorted the images into the two groups based on their manufacturer tag in the DICOM header: the 'Varian Medical Systems' tag for CBCT scans, and the 'Philips' tag for CT scans.

We converted the images to NIFTI format using the `dicom2nifti` library.

In the dataset, each CBCT image has been registered to the planning CT and then resampled to match the voxel dimensions and volume of the planning CT. Thus, CBCT/CT images from the same patient have same voxel dimensions, but the voxel dimensions of the images vary among patients. The voxel size of an image was in-

ferred from the Pixel Spacing attribute (first two dimensions) and the Slice Thickness attribute (third dimension) in the DICOM header. The voxel dimensions vary between 1.37 x 1.37 x 3.0 mm^3 and 0.98 x 0.98 x 1.5 mm^3. In comparison, all images of the BraTS2020 dataset have the same voxel dimensions.

It is usually preferred to have equal spatial resolution/scale / voxel size in the training images input to a CNN, since this can facilitate training and boost performance [Van Noord and Postma, 2017]. Standard single-scale CNNs trained without data augmentation are scale-variant or scale-specific, which means that if the scale in input changes, the output changes. They learn task-relevant features at a specific scale and can't handle scale variants of the same features at different scales. If a CNN is exposed to multiple input scales, the convolutional filters should work for all scales, but this may mean that it fails to learn important task-relevant information and thus will produce less precise outputs. Whereas if the input scale is constant, the filters can specialise at this specific scale, which improves the quality of generated images.

Another thing to consider is alignment of images. Image registration can be used to align images in the training dataset to make them comparable [Masoudi et al., 2021] i.e. by constructing the transformation between the images of different modalities that maps similar features such as anatomical structures correctly [Leng et al., 2013]. Image registration can be used to obtain equal voxel spacing in the images.

Because the voxel sizes in our dataset do not vary significantly, we chose not to do anything with the images. Most importantly is that the voxel dimensions among the pairs of CBCT/CT images are the same, because then the network will see more of the same anatomy from the different modalities.

3.2 Training and memory restrictions

Since we have limited memory capacity, we cannot just train our models with the original volumes as input. We have mainly been working on notebooks in Google Colab, and here the most suitable input volume size to the network has been e.g. $32 \times 32 \times 32$ or $8 \times 64 \times 64$ (depth $\times$ height$\times$ width). We chose to mainly work with the last volume size as that gave the most easily displayable volumes during training with enough detail to get an idea of what was happening. On Google Colab, this limited our training to batch sizes of 4.

There are two main ways to shrink the size of the input volumes. The first is to downsample the whole image to the desired size, the second to extract patches of the desired size from the whole image. The first method will be able to learn information from the whole image, but will not learn the details very well. The second method will be able to learn details well, but will not learn from the whole image. In the experiment section further below, we experiment with both these methods and (hopefully) also show the shortcomings of each.

3.3 Data preprocessing

The purpose of the model is to be able to be trained unsupervised on unpaired data. Both the BraTS 2020 and the Pancreatic-CT-CBCT-SEG are paired datasets, so in order to simulate an unpaired data set we shuffled the data path lists so our dataloader would not load paired sets.

In order to feed our data to the neural networks of our model, we need to normalize the voxel values. To keep it simple, we use min-max-normalization for both MRI and CT volume, with each volume determining its own minimum and maxmium value.

For training on downsampled images, we simply use the `skimage.transform.resize`-function to downsample the images to the desired size.

When training on patches, we go through each volume as it is loaded by the dataloader and extract a random patch. Initially, this created problems as there is a lot of black in the original volumes. The randomly chosen patches would often be black, and since the batch size is very small (2 or 4), it would only take for one or two of the random patches to be black before the generators started exclusively producing black images. For this reason, we had to first crop the images to remove as much black as possible. This was simple for the MRI scans, as they had the same size and the same field of view so we could crop them all in the same manner. For the CT and CBCT scans it was more difficult, as they all had different sizes and as the field of view of the CBCT scans were often much smaller than the field of view of the CT scans, meaning there was a lot more black in the CBCT scans than in the CT scans. Our solution for this ended up being determining for each CBCT scan how to crop it, and use the same crop for the CT scan that was loaded alongside it.

3.4 Model

3.4.1 Discriminator

For the discriminators, we used a modified version of the implementation of the PatchGAN discriminator (specifically called the `NLayerDiscriminator`) provided in the GitHub repository of the authors of the original CycleGAN paper [Zhu and Park, 2017]. It consists of 5 convolutional layers, using LeakyReLU for non-linearity and layer normalization after the second, third and fourth layer. The original PatchGAN discriminator was meant for 2d images, so we modified it by exchanging the 2d convolutions with 3d convolutions. In the original PatchGAN, the kernel size used was 4×4, but to allow the discriminator to handle smaller volume sizes, we had to change the kernel size to $3 \times 3 \times 3$. The original PatchGAN classifies patches of 70×70 of the original input as either true or false [Isola et al., 2016]. In our modified PatchGAN, when the input is of size e.g. $8 \times 64 \times 64$, patches of $8 \times 45 \times 45$ of the original input are classified as true or false,

i.e. one pixel in the output activation map has a receptive field of $8 \times 45 \times 45$.

3.4.2 Generator

For the generators, we used almost verbatim the implementation of a 3d Unet originally created by Abraham George Smith [Smith, 2020] for segmentation purposes (as of the time of writing this report, this repository is unfortunately no longer visible on GitHub). It was slightly modified to suit our memory restrictions by reducing the number of downsampling/upsampling steps by one, and by changing the number of input and output channels from 3 to 1.

3.4.3 CycleGAN

The implementation of the actual model, the CycleGAN, is based on a simplified version of the CycleGAN implementation provided in the GitHub repository of the authors of the original CycleGAN paper [Zhu and Park, 2017].

In the forward pass, the generators create the fake A and B images from the real B and A images, respectively, as well as the reconstructed A and B images from the fake A and B images. We then have two separate backward passes, one for updating the gradients of the discriminators and one for updating the gradients of the generators.

We recall that the discriminators must try to maximize the adversarial loss function described in 2.5. The output of the discriminator is a set of probabilities that patches of an image are either real or fake. To maximize the adversarial loss function is therefore to minimize the distances between the probabilities and the real classifications, i.e. to let the discriminator strive to learn to predict correctly whether an image is real or fake. This is achieved in the backward pass for the discriminators using Binary Cross Entropy as the loss function measuring the distance between the predictions and the real classifications, where the real classifications are represented by either a tensor of ones or a tensor of zeros.

The generators must try to minimize the adversarial loss function. This is done by learning to generate images that are of good enough quality to fool the discriminator into classifying them as real. This is achieved in the backward pass for the generators using Binary Cross Entropy as the loss function measuring the distance between the discriminators' predictions for the fake images and the classification that the generator would like the fake images to have, represented by tensors of ones. It is also in the backward pass for the generators that the cycle losses are calculated - i.e. the differences between the real images and the reconstructed images - using the L1 loss function.

After the backward pass of the discriminator, the gradients for the discriminator networks are updated using the Adam optimizer according to the sum of the losses calculated for the discriminators. Likewise, after the backward pass of the genera-

tor, the gradients of the generator networks are updated using the Adam optimizer according to the sum of the losses calculated for the generators.

Initial training experiments suggested that the best learning rate was around 0.0001. When the learning rate was lower, the model learned much more slowly, and if it was larger the resulting images rarely improved in quality past a certain early point. We also tested different weights λ for the cycle-consistency loss, but did not see any big differences, so in the end we just stuck with $\lambda = 10$, as was also done for the original CycleGAN paper [Zhu and Park, 2017].

3.5 Evaluation

As of now, the question of how to best evaluate generative models has not been settled, but the most common way to do it is to evaluate the quality and diversity of the synthetic images. While we can do qualitative manual evaluation by inspecting the generated images, there are pitfalls to this method. The evaluator is influenced by own biases and, when it comes to our domain, medical images, must have an extensive knowledge of the subject. There is also a natural limit as to how many images can be inspected by humans. For these reasons, we also include one of the most popular quantitative metrics for evaluating generative models, the Fréchet Inception Distance score, to complement the manual evaluation. The purpose of the Fréchet Inception Distance score is to compare the statistics of one set of images (the real targets) to the statistics of another (the generated targets). It is supposed to be consistent with human inspection and sensitive to mode collapse and to small changes in the real distribution. Drawbacks include the assumption of a Gaussian distribution and the need for large sets of images (10000+) to avoid overestimating the real FID score.[Borji, 2021]

3.5.1 FID score

To calculate the FID score for a set of real and generated images, a pretrained Inception v3 model is used. The Inception v3 model is a model for classifying images. The activation map provided by the last pooling layer in the model is extracted for both the set of real and the set of generated images. For each activation map, the mean and the covariance is calculated and two multivariate Gaussians are determined from these. The Fréchet Inception Distance score is then calculated as the Wasserstein-2 distance between these two distributions. The lower this score is, the better the quality of the generated images.

The FID score is calculated for 2d images. As we are working with 3d images, we calculate the FID score based on the slices of the volumes. A relatively well-known pytorch implementation for calculating FID scores of images saved in folders is developed by [Seitzer, 2020]. In order to calculate the FID scores in memory, we use a wrapper for the original implementation by [Schmidt, 2020].

In the experimental phase, we do not have access to 10000+ images, but as we

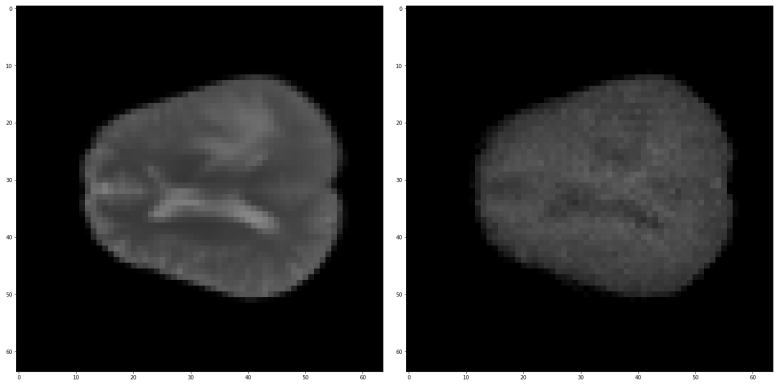
mainly use the FID score to track how well our model is learning during training and not to compare the actual scores with the scores of other models, this should not be a big issue.

4 Experiments

4.1 MRI

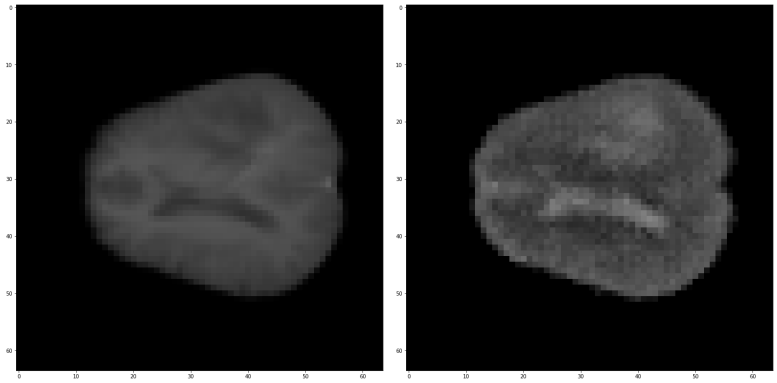
4.1.1 Overfitting

To get an idea of whether our implemented CycleGAN was able to learn as well as get a feeling for whether our FID score calculations made sense, our first experiments consisted of training the CycleGAN on only one patient in order to get it to overfit. We downsampled the paired T2 and T1ce volumes of one patient to $8\times64\times64$ volumes and let the CycleGAN train on these for 200 epochs. In Figure 3, we see examples of the translations the CycleGAN was able to make after 200 epochs of training. The slice from a translation of the T2 volume to the T1ce modality does indeed look like it is getting close to look like the corresponding slice in the original T1ce volume with the slightly altered overall shape of the brain and the darkening of the ventricles. Likewise, the slice from a translation of the T1ce volume to the T2 modality looks close to the corresponding slice of the original T2 volume, again with the slightly altered overall shape and the brightening of the ventricles.



(a) A slice from the real T2 volume

(b) Corresponding slice from fake T1ce volume generated from the real T2 volume



(c) A slice from the real T1ce volume

(d) Corresponding slice from fake T2 volume generated from the real T1ce volume

Figure 3: Overfitting on downsampled T2 and T1ce volumes from a single patient

When looking at the FID score throughout the training (Figure 4), we also see a clear downward trend, indicating that the distribution of the generated T1ce images becomes more and more like the distribution of the original T1ce images.

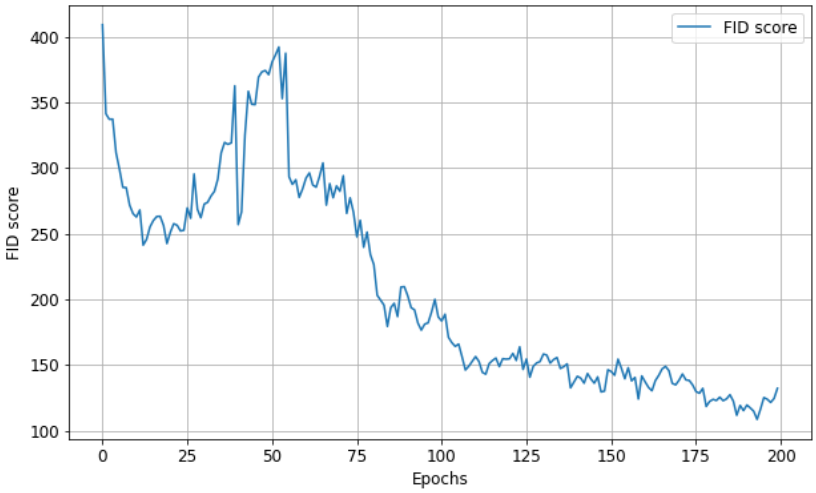


Figure 4: FID score for generated and real T1ce volumes during training on one patient

Finally, in Figures 5 and 6, we see the losses of the two GANs that the CycleGAN is made up of. In the cases of both GANs, we see that when the adversarial loss for the discriminator (D_B and D_A, respectively) decreases, the adversarial loss for the generator (G_AtoB and G_BtoA, respectively) increases, and vice versa. This is as we would expect: if the discriminator is a bit better at judging real and false volumes it means that the generator is a little bit worse at creating convincing false volumes, meaning the discriminator loss goes down and the generator loss goes up. As the task of generating convincing volumes is a more complicated task than discriminating between real and fake volumes, it is also no surprise that the generator loss is much higher than the discriminator loss in the beginning, before they both stabilize from around the 100th epoch.

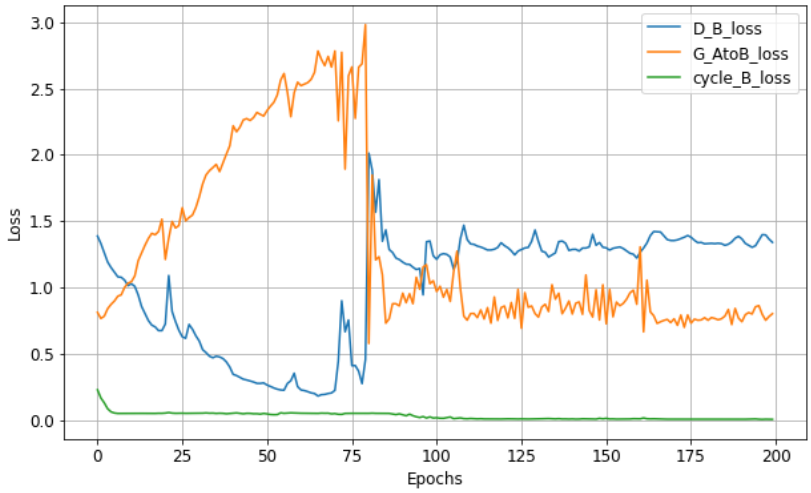


Figure 5: Loss values for the GAN translating from A to B (T2 → T1ce)

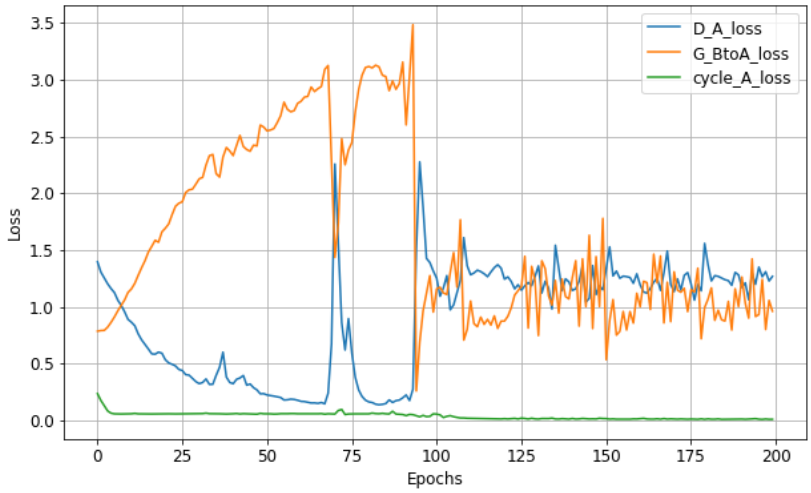


Figure 6: Loss values for the GAN translating from B to A (T1ce → T2)

4.2 CT

4.2.1 Training on downsampled volumes

We proceeded to split the Pancreatic-CT-CBCT-SEG data set up into a training set of 36 patients (we shuffled the patients to simulate an unpaired dataset) and a test

set of 4 paired patients. Since there were two CBCT volumes for each patient and there did not seem to be any significant differences between them, we chose one of those two CBCT volumes at random.

We then trained an instance of our CycleGAN model on the downsampled volumes in the training set for 300 epochs. The FID scores calculated along the way can be seen in Figure 7. After a quick decrease of the FID scores followed by a longer set of increasing FID scores, the score generally starts decreasing at a slow but consistent rate from around the 60-70th cycle. If we had trained for more epochs, we would have expected the scores to go even lower.

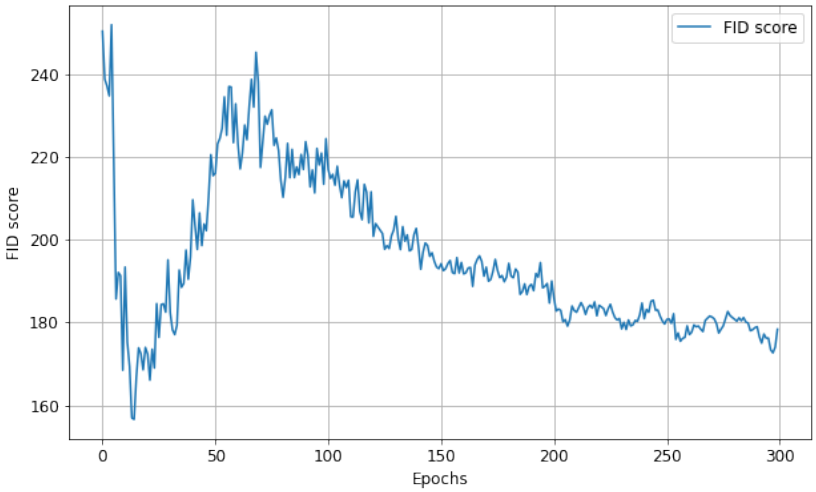


Figure 7: FID scores for generated and real CT volumes volumes during training with downsampled volumes

In Figures 8 and 9, we see the plots for the loss values for the two GANs that make up the CycleGAN. In both cases, while the generator and the discriminator losses still reflect each other in that one goes down when the other goes up, they are fluctuating quite a lot and moving far away from each other, which might be a sign that it has a difficult time learning. After all, the CBCT volumes often have much smaller fields of view and the amount of slices with actual information on them are often much fewer than in teh CT volumes, so it is not surprising that it would be hard to learn.

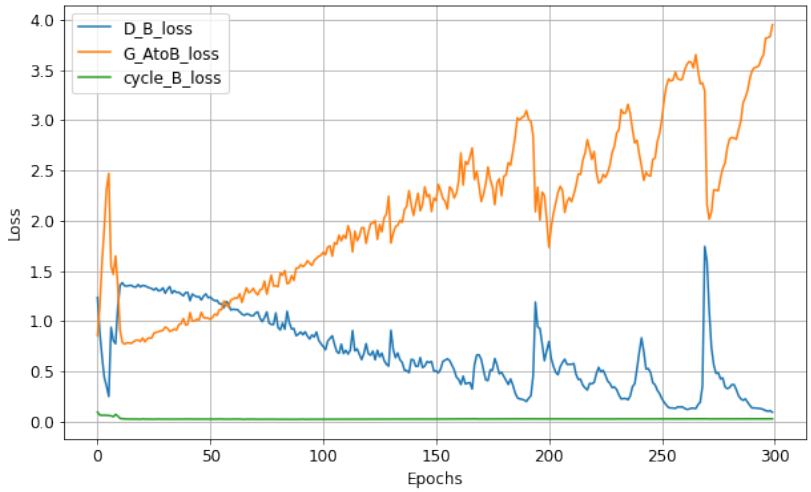


Figure 8: Loss values for the GAN translating from CBCT to CT (trained on down-sampled volumes

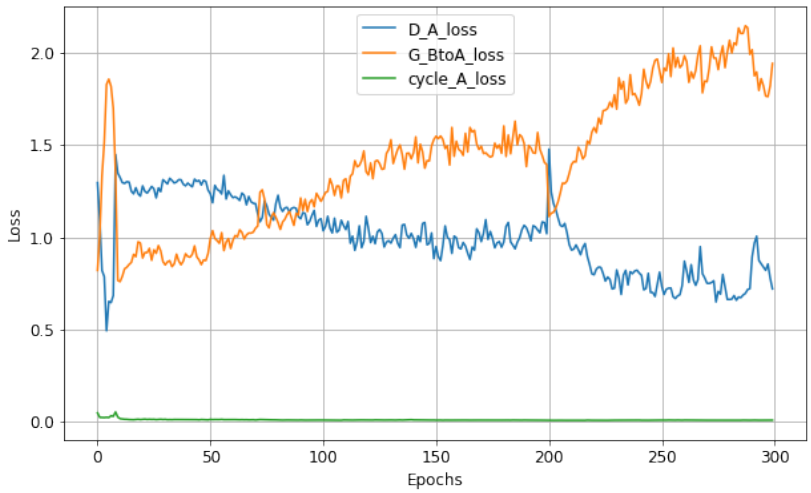
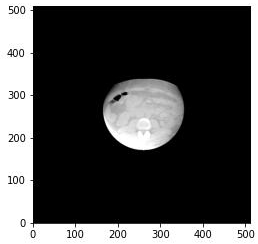


Figure 9: Loss values for the GAN translating from CT to CBCT (trained on down-sampled volumes

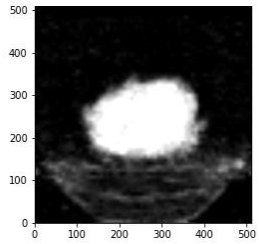
After training the CycleGAN for the 300 epochs, we tested its ability to translate an original CBCT volume to a CT volume, and vice versa. Examples of the results are seen in Figures 10 and 11, respectively. The original volume was normalized,

downsampled, translated, upsampled, and then denormalized.

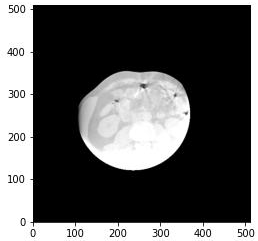
As previously mentioned, the fields of view of the CBCT volumes are generally much smaller than those of the CT volumes, while the CT volumes often have such a big field of view that the bed the patient is lying on is also visible in the scans. In Figure 10, we clearly see that, although the example slices of the translated volumes are of very blurry quality, they have learned to fill out the generated CT images so that their field of view appears larger. Likewise, in Figure 11, we see that the model has also learned that the CBCT volumes of our training set ususally have smaller fields of view.



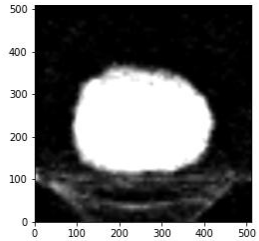
(a) A slice from an original CBCT volume (test patient 0)



(b) Corresponding slice from fake CT volume generated from the original CBCT volume (test patient 0)

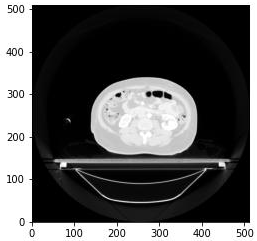


(c) A slice from an original CBCT volume (test patient 2)

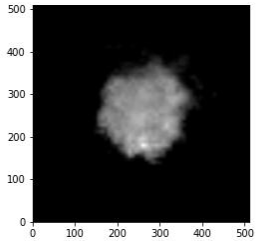


(d) Corresponding slice from fake CT volume generated from the original CBCT volume (test patient 2)

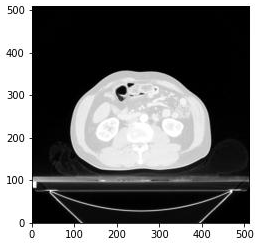
Figure 10: Examples of translations using the generator CBCT->CT trained on downsampled volumes



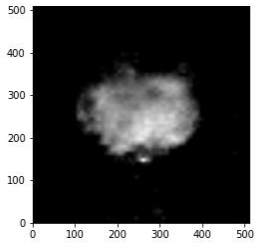
(a) A slice from an original CT volume (test patient 0)



(b) Corresponding slice from fake CBCT volume generated from the original CT volume (test patient 0)



(c) A slice from an original CT volume (test patient 2)



(d) Corresponding slice from fake CBCT volume generated from the original CT volume (test patient 2)

Figure 11: Examples of translations using the generator CT->CBCT trained on downsampled volumes

4.2.2 Patch training

As explained in 3.3, in order for our patch training to succeed and not just end up with black images, we had to severely crop our volumes before we extracted the random patches. We then trained an instance of our CycleGAN on randomly extracted patches for 66 epochs. We would have like to train for longer, but for various reasons did not succeed in this. However, it is still possible to get a feeling for the quality of the training. In Figure 12, we see that the FID scores for the patch training in general have a downward decreasing tendency. Likewise, in the loss plots in Figures 13 and 14, we see that the loss values for the discriminator and the generator do not fluctuate a lot.

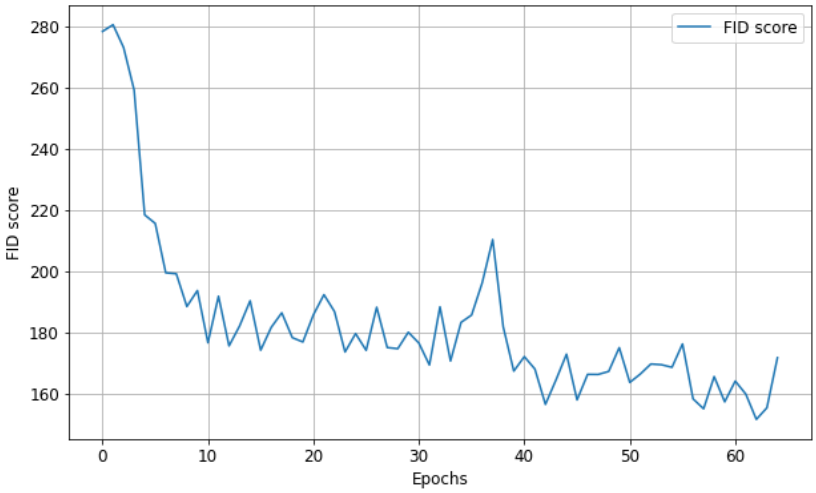


Figure 12: FID score for generated and real CT patch volumes during patch training

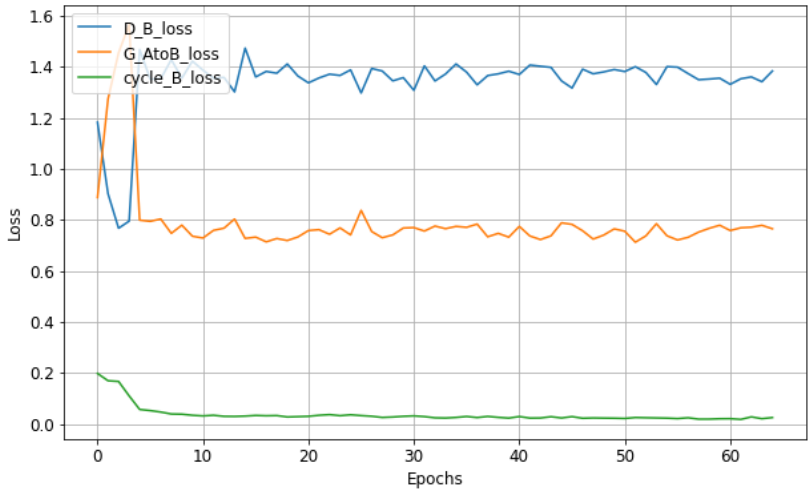


Figure 13: Loss values for the GAN translating from CBCT to CT (patch training)

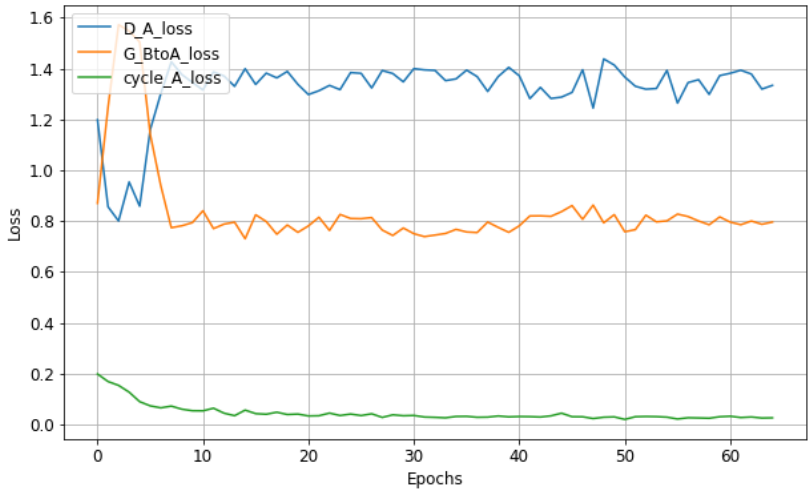
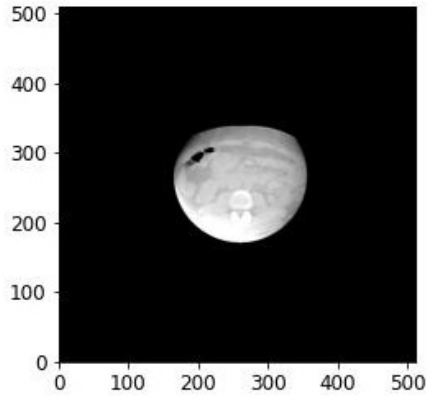


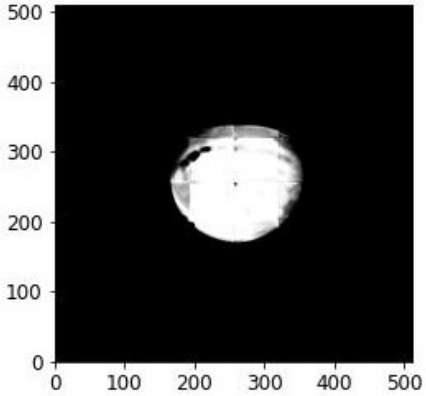
Figure 14: Loss values for the GAN translating from CT to CBCT (patch training)

After training the CycleGAN for the 66 epochs on the patches, we tested its ability to translate an original CBCT volume to a CT volume, and vice versa. Examples of the results are seen in Figures 15 and 16. The original volume was split into patches of the same size that the CycleGAN was trained on, i.e. in this case patches of size $8 \times 64 \times 64$. They were split in such a way that there was the least possible overlap. If our purpose here had been to actually create the best possible translation with these patches we might have taken care to make them overlap much more. However, for illustrative purposes we did it with less overlap. Each patch was then translated using the proper generator of the CycleGAN and the translated patches assembled together to form a volume of the original shape. Overlaps where handled by taking the average of the voxel intensity.

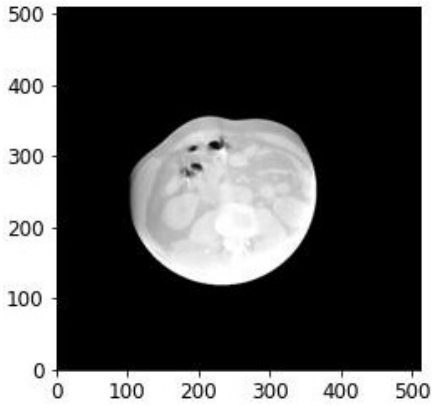
As we can can see, the quality of the example slices of the generated volumes are of much sharper quality than when translating with the CycleGAN trained on downsampled volumes. However, we also see that they do not learn as well from the bigger picture. The original CT volumes often have the bed in the field of view, but this has not been learned here, as can be seen in Figure 15. Furthermore, we also see that the generated images are not of continuous quality as the patches do not blend seamlessly with each other.



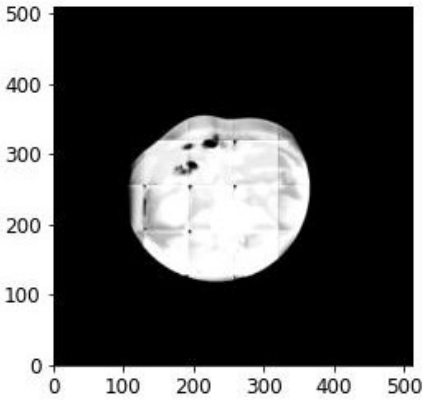
(a) A slice from an original CBCT volume (test patient 0)



(b) Corresponding slice from fake CT volume generated from the original CBCT volume (test patient 0)

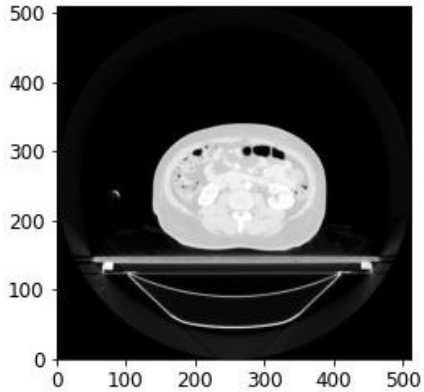


(c) A slice from an original CBCT volume (test patient 2)

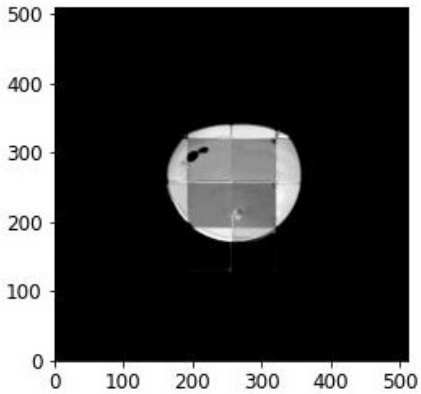


(d) Corresponding slice from fake CT volume generated from the original CBCT volume (test patient 2)

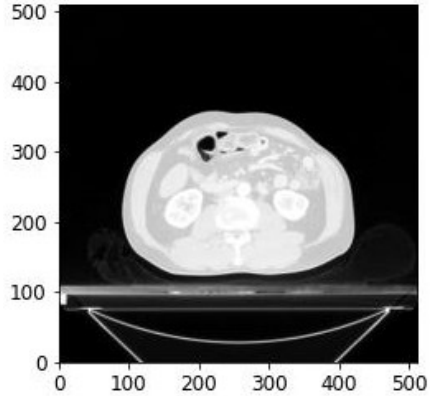
Figure 15: Examples of patch translations using the generator CBCT->CT trained on patches



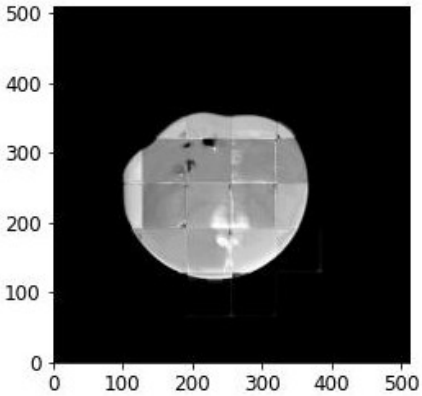
(a) A slice from an original CT volume (test patient 0)



(b) Corresponding slice from fake CBCT volume generated from the original CT volume (test patient 0)



(c) A slice from an original CT volume (test patient 2)



(d) Corresponding slice from fake CBCT volume generated from the original CT volume (test patient 2)

Figure 16: Examples of patch translations using the generator CT->CBCT trained on patches

5 Discussion

With our two CycleGANs trained on patches and downsampled volumes, respectively, we were able to demonstrate the strengths and weaknesses of each approach, although the actual quality of the translated volumes made by the generators of these CycleGANs was not very high. There would have been several ways to improve on this. The Pancreatic-CT-CBCT-SEG data set that was used to train the two CycleGANs consisted of only 40 patients, 4 of which were reserved for testing. If we had gotten as far as using the much more plentiful data that Rigshospitalet would have been able to provide, this would probably have been grounds for improvement. As we saw, the FID scores had a general tendency to decrease

after a certain point in training, so we would also expect that training for a longer time would have improved the quality of the translations provided by each of the individual CycleGANs, even without the added data.

Seeing as the CycleGAN trained on downsampled volumes and the CycleGAN trained on patches each brought something to the table in terms of the quality of the translations, one way to improve the translation model could be to combine these two aspects in one model. A simpler way to improve the quality of the translation could also have been to let a more well-trained patch CycleGAN translate patches with high overlap. This could have the effect of smoothing over the edges between the patches that was otherwise very visible in our demonstrations, but would probably still pose problems in terms of learning information from the big picture.

6 Conclusion

We have conducted a literature review of medical image synthesis with the purpose of finding a method to translate between CBCT and CT images. Generative adversarial networks (GANs) were given the pride of place, since these are considered the state-of-the-art in the field. Our focus was on cross-modality image synthesis with conditional GANs since this related mostly to our problem. Studies showed that the CycleGAN model, which originally was proposed for 2d images, can be used to successfully translate between different image modalities, including CBCT and CT. CycleGAN has proven to out-perform other GAN models, e.g. DCGAN for CBCT to CT translation, while the vanilla CycleGAN itself has been improved by addition of non-adversarial losses in e.g. Cycle-MedGAN, structure-constrained (sc) CycleGAN and gradient-consistency (gc) CycleGAN. The additional losses gave a more precise translation with respect to details and preserved anatomy better in the synthesized images. Furthermore, 3d CycleGANs have shown even better results than basic 2d CycleGANs, since they are better to model the spatial information in the 3d medical volumes. However, 3d models are more difficult to train due to the larger GPU memory requirements, which can be coped with in different ways (downsampled images, patches, etc.), but unfortunately these approaches introduce new problems instead. Other 3d GANs have also showed good results, such as a cascaded context-aware GAN, but is less applicable in practice since it relies on paired image sets, which are difficult to obtain. In our literature review we found that generally it helps to have paired data (if one is that lucky), and the more supervised the translation method, usually the better the quality of the generated images. However, when not in possession of paired data, ground-truth segmentations or other labels to supervise the translation, CycleGAN is a great choice and mostly shows very reasonable results with unpaired data. Registration as a pre-processing step can improve the performance of unsupervised synthesis methods.

We implemented a 3d CycleGAN to translate between CBCT and CT images. At first, when we only had access to the BraTS 2020 MRI data set, we used this to demonstrate that our CycleGAN was actually capable of learning by overfitting the model on the T2 and T1ce volumes of a single patient. As we got access to the Pancreatic-CT-CBCT-SEG data set, we had to adjust our data preprocessing as the differences between the CT and CBCT volumes were much greater than those between the T2 and T1ce MRI volumes. We then proceeded to train two CycleGANs using the Pancreatic-CT-CBCT-SEG data set. The first was trained on downsampled volumes for 300 epochs. Using the generators of this CycleGAN to translate original volumes, we demonstrated that a CycleGAN trained on downsampled volumes was able to learn information on a big scale at the cost of producing images of a very blurry quality. The second CycleGAN was trained on randomly extracted patches from the original volumes. Using the generators of this CycleGAN to translate original volumes, we demonstrated that a CycleGAN trained on patches is able to produce images of a much sharper quality and with much more detail, at the cost of non-continuity between patches and of not learning information on a larger scale. While none of the trained models were producing translations of good enough quality to use as independent checks for the Adaptive Radiation Treatment system at Rigshospitalet, there are many ways to improve upon this. Training for longer times and using the bigger and better data sets at Rigshospitalet is one such way, another is to modify the model to incorporate training on both patches and downsampled volumes.

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